Week 19: Learn from Time, Generate Better Research Ideas

Time-Series Modeling and a Week 18 Research-Idea Workshop

Instructor / Mentor

Kiki Research Training Program

Week 19 of the Revised 22-Week Program

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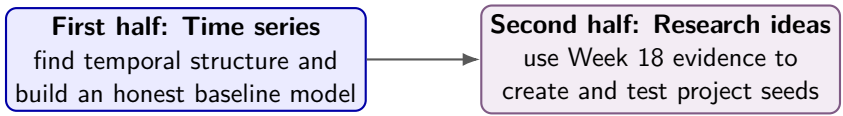
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Two Missions for Today

Today's Two Missions



Shared habit

Respect order and evidence. In a time series, order carries information. In research ideation, evidence should come before a confident claim.

Learning Objectives

By the end of class, students can:

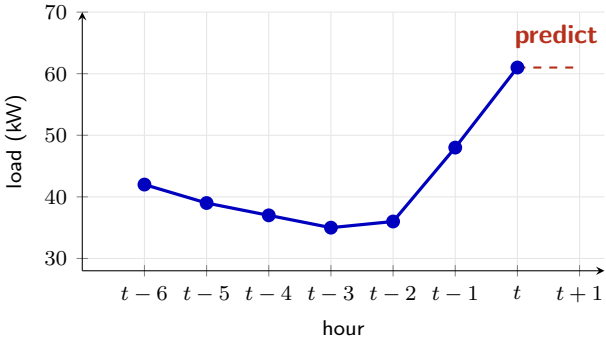
1. recognize trend, seasonality, events, and noise in a time plot;
2. create lag and rolling features without using future information;
3. explain why time-series data require a chronological train/test split;
4. compare a simple model with persistence and seasonal-naive baselines;
5. inspect forecast errors instead of reporting only one score;
6. extract reusable research moves from the Week 18 materials;
7. generate several distinct, testable research ideas; and
8. choose a primary idea and a backup using evidence and feasibility.

120-Minute Flow

1. **0–8 min:** opening forecast challenge and define a time series.
2. **8–24 min:** visualize trend, seasonality, events, and missingness.
3. **24–39 min:** lags, rolling summaries, and leakage prevention.
4. **39–54 min:** baselines, chronological evaluation, and residuals.
5. **54–60 min:** mini-model checkpoint and break.
6. **60–72 min:** unpack the Week 18 homework evidence.
7. **72–91 min:** use four idea generators to produce six seeds.
8. **91–106 min:** convert seeds into testable research questions.
9. **106–116 min:** feasibility tournament and mentor feedback.
10. **116–120 min:** choose the next evidence check and launch homework.

Time-Series Analysis and Modeling

Opening Challenge: What Happens Next?



60-second forecast

Write one number for $t+1$, then name the pattern or assumption behind it.

You might use:

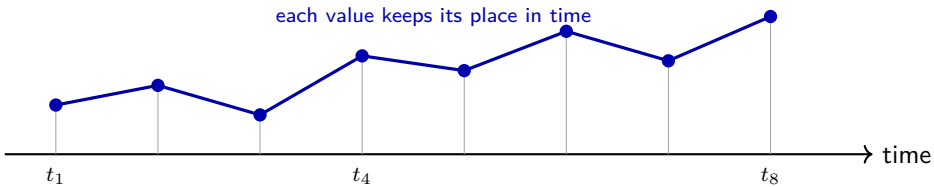
- the latest value;
- a local trend;
- the same hour yesterday;
- weather or occupancy.

Forecasts are conditional arguments

The number is only as meaningful as the information, time horizon, and assumptions behind it.

What Makes a Dataset a Time Series?

A time series is a sequence of observations attached to ordered times: $\{(t_i, y_i)\}_{i=1}^{n}$.



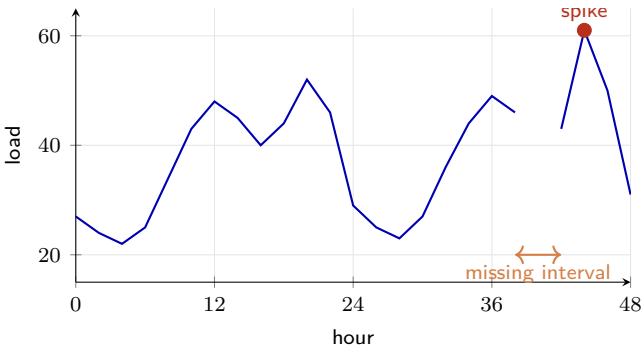
Examples

- hourly electricity load;
- minute-by-minute frequency;
- daily solar generation.

Why order matters

Shuffling the rows destroys recency, cycles, event order, and the distinction between past and future.

Begin with the Time Plot, Not the Model



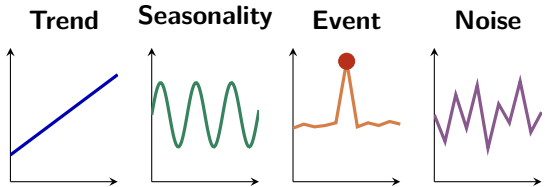
Look before fitting:

1. sorted and unique?
2. regular interval?
3. gaps or impossible values?
4. changing level?
5. repeating pattern?
6. meaningful events?

Research habit

A strange point can be noise, an error, or the most important event in the dataset. Investigate before deleting it.

Four Things We Often See in Time



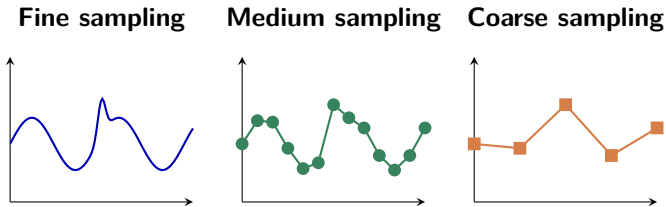
A useful visual and mathematical checklist is

$$y_t \approx \text{level/trend}_t + \text{seasonal pattern}_t + \text{event effects}_t + \text{noise}_t.$$

These pieces can overlap

The decomposition guides questions; real series do not always separate perfectly into four independent parts.

Sampling Frequency Changes the Question



Fault recovery in seconds?

Solar ramp in 15 minutes?

Peak demand tomorrow?

Do not choose the horizon after seeing the answer

Sampling can reveal or hide an event. Define what is predicted, how far ahead, and for whom before comparing models.

Missing Time Is Not the Same as a Zero Value



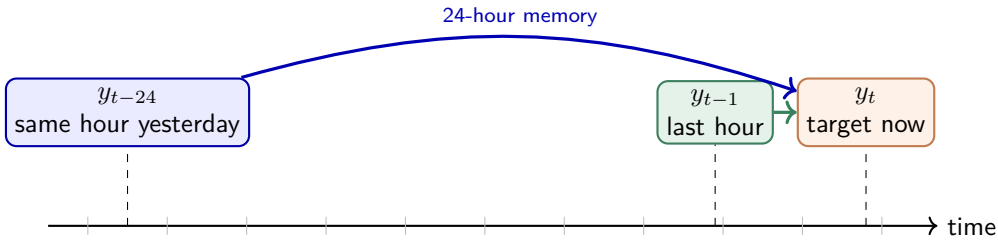
Keep an audit trail

Preserve `was_missing`. Interpolation creates an estimate, not a recovered observation.

The correct treatment depends on why data are missing and how the result will be used.

A Lag Feature Gives the Model Memory

A lag aligns a past observation with today's target: $\text{lag}_k(t) = y_{t-k}$.

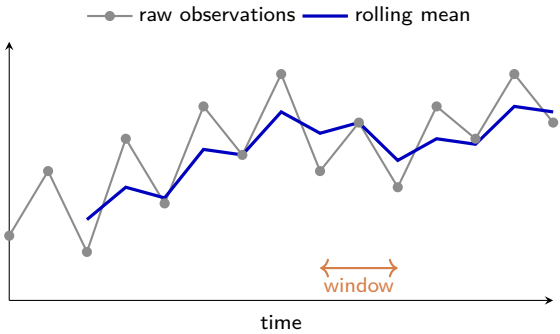


y_{t-168} can add "same hour last week" memory when weekly behavior matters.

Direction matters

A negative shift uses future information. Features for predicting y_t may use y_{t-1}, but never y_{t+1}.

Rolling Summaries Describe Recent Context



$$m_t = \frac{1}{w}\sum_{i=1}^{w} y_{t-i}$$

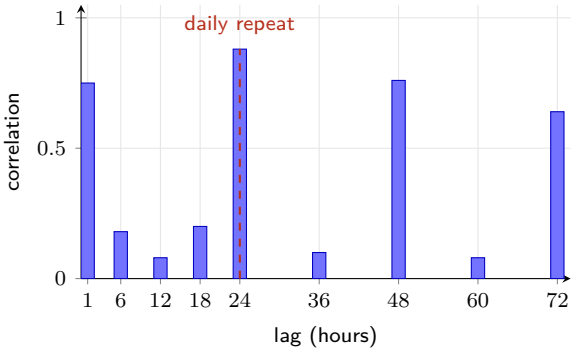
Rolling summaries show:

- recent level;
- recent volatility;
- recent extremes;
- recent movement.

Safe construction

Shift first, then roll: `load.shift(1).rolling(24).mean()`. This excludes the target value from its own feature.

Autocorrelation Asks Whether the Past Resembles the Present



A peak near lag 24

An hourly series may repeat a daily pattern, so the same hour yesterday is informative.

What it does not prove

Correlation does not establish causation, guarantee future stability, or explain why the cycle exists.

Domain knowledge should guide which lags are tested; searching many lags can find accidental patterns.

Define the Forecasting Task Before Training

A complete task states five choices:

Unit
one building

Target
load in kW

Horizon
one hour ahead

Inputs
available by t

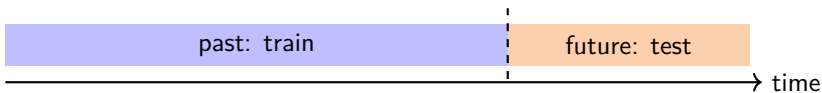
Metric
MAE and plots

Information boundary

Imagine freezing the clock at prediction time. Every feature must already be known at that moment.

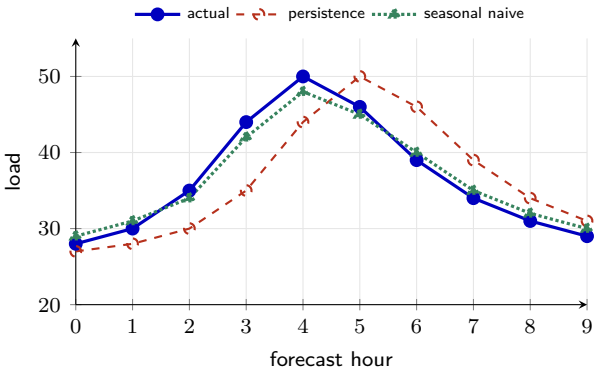
Random Splitting Leaks the Future

In ordinary tabular learning, rows may be shuffled. In forecasting, later observations must not train a model evaluated on earlier ones.



Fit scalers, imputers, feature choices, and model parameters using the training period only. For repeated evaluation, move the cutoff forward in time.

Always Challenge the Model with Simple Baselines



Persistence: $\hat{y}_t = y_{t-1}$

Seasonal: $\hat{y}_t = y_{t-s}$

Mean: $\hat{y}_t = \bar{y}_{\text{train}}$

Why baselines matter

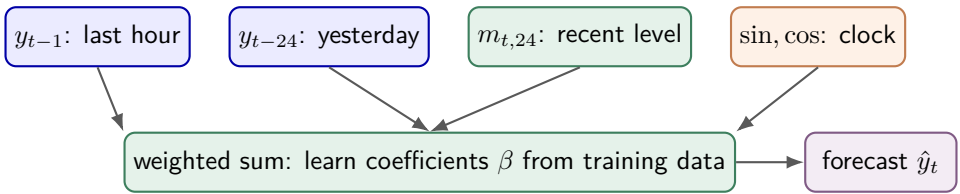
A complex model should beat a reasonable simple rule on future data.

For hourly building load, $s = 24$ may be sensible; weekly behavior may suggest $s = 168$.

A Simple Lag-Feature Regression

A first transparent model combines past observations and calendar signals:

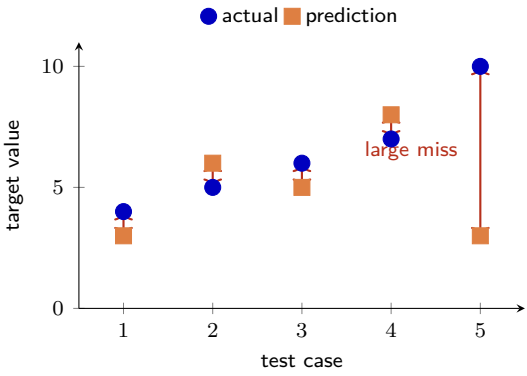
$$\hat{y}_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-24} + \beta_3 m_{t,24} + \beta_4 \sin(2\pi h_t/24) + \beta_5 \cos(2\pi h_t/24).$$



Transparent does not mean automatically correct

The learned coefficients are inspectable, but future-data performance must still beat the baselines.

Measure Error in More Than One Way



$$\text{MAE} = \frac{1}{n}\sum |y_i - \hat{y}_i|$$

MAE

Typical absolute miss in the target's units; comparatively easy to explain.

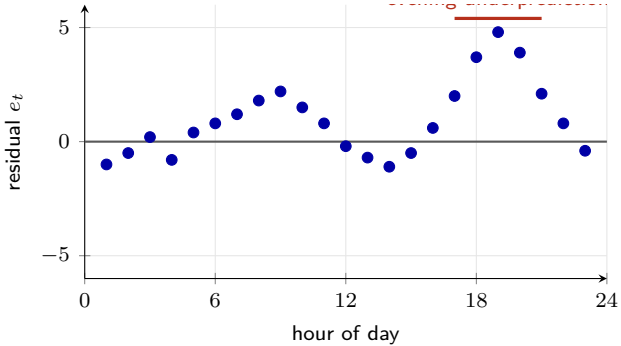
$$\text{RMSE} = \sqrt{\frac{1}{n}\sum (y_i - \hat{y}_i)^2}$$

RMSE

Penalizes large misses more strongly; useful when large errors are especially costly.

MAPE can become unstable or undefined when actual values are zero or close to zero.

Residuals Tell Us What the Score Hides



$$e_t = y_t - \hat{y}_t$$

Ask whether errors:

- center near zero;
- grow over time;
- repeat by hour or week;
- worsen during peaks;
- differ across groups.

A research clue

A pattern in residuals is not just a lower score—it points to information or behavior the model has not explained.

Mini Pipeline: Past Information to Future Evaluation

```python
df = df.sort_values("time").set_index("time")
df["lag_1"] = df["load"].shift(1)
df["lag_24"] = df["load"].shift(24)
df["mean_24"] = df["load"].shift(1).rolling(24).mean()
df = df.dropna()

cut = int(0.8 * len(df))
train, test = df.iloc[:cut], df.iloc[cut:]

features = ["lag_1", "lag_24", "mean_24"]
X_train = np.column_stack([np.ones(len(train)), train[features]])
beta = np.linalg.lstsq(X_train, train["load"], rcond=None)[0]
X_test = np.column_stack([np.ones(len(test)), test[features]])
test["model_pred"] = X_test @ beta
test["persistence"] = test["lag_1"]
test["seasonal_naive"] = test["lag_24"]
```

Checkpoint

Point to the line that protects temporal order, the lines that create memory, and the predictions that act as baselines.

First-Half Checkpoint

In pairs, answer without notes:

1. Why is a random split dangerous for forecasting?
2. Why should a rolling feature usually be shifted before rolling?
3. What is the difference between persistence and seasonal naive?
4. If model MAE is lower but peak-hour errors are much worse, is it clearly the better model?
5. Name one residual pattern that could become a research question.

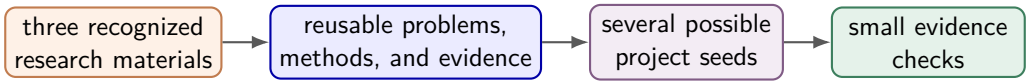
Six-minute break

Bring the completed Week 18 Research Inspiration Summary back with you.

From Week 18 Evidence to Research Ideas

Restart with Evidence, Not a Blank Page

Place the Week 18 summary beside your Week 17 proposal seed.



Today's goal

Do not defend the first idea. Create alternatives, expose assumptions, and leave with a primary idea plus a credible backup.

Fast Homework Debrief: One Insight per Material

For each of the three materials, write one sentence in each column:

Material	Strong research move	Important limitation or uncertainty	Idea it changes or creates
1			
2			
3			

Cite the sentence, paragraph, figure, or table that supports each claimed move. If support is incomplete, label it as an inference or question.

Borrow a Research Move, Not a Research Identity

Inspiration does not mean copying a title and changing the location.

Reusable moves

- compare against a meaningful baseline;
- measure something hidden by an average;
- join technical and human outcomes;
- release a useful dataset or tool;
- test a method under a failure condition.

Weak imitation

- reuse the same claim without access to the evidence;
- add "AI" without a decision or comparison;
- assume a new community has the same needs;
- claim impact without a pathway.

Idea Generator 1: Transfer

Move a method or measurement into a new but justified context.

Frame

Material [A] used *[method/measurement]* to study *[problem]*. Would it reveal something useful about *[new context]*, and what would need to change?

Examples of transfers:

- a power-quality metric from neighborhoods to school buildings;
- disturbance detection from a simulated grid to an open sensor dataset;
- a stakeholder interview framework from households to facility managers.

A transfer needs a reason the contexts are comparable and a plan for their differences.

Idea Generator 2: Compare

Turn an uncertain choice into an experiment.

Frame

For *[outcome]*, how does *[method or group A]* compare with *[method or group B]* under *[same conditions]*?

Time-series examples:

- persistence versus seasonal naive for peak-load forecasting;
- hourly versus daily aggregation for detecting outages;
- overall MAE versus subgroup or peak-period error;
- a model trained on normal days versus one evaluated during extreme days.

A fair comparison fixes the target, horizon, test period, and metric.

Idea Generator 3: Expose a Hidden Assumption

Find a statement that must be true for the original method or impact to work.

Frame

The project assumes *[condition]*. What happens to *[outcome]* when that condition fails or differs across groups?

Candidate assumptions include:

- sensors report regularly and accurately;
- yesterday resembles today;
- training data represent future operating conditions;
- a technical improvement benefits all users similarly;
- the chosen metric captures what stakeholders value.

Idea Generator 4: Build a Small Enabling Resource

Sometimes the contribution is a tool that makes later research possible.

Frame

Researchers or stakeholders cannot easily *[task]*. I will create and evaluate a small *[dataset, visualization, checklist, or tool]* that helps them *[decision]*.

Examples:

- a reproducible missing-data audit for smart-meter files;
- a dashboard comparing forecast errors across hours or neighborhoods;
- a labeled collection of grid events with clear definitions;
- a guide matching research questions to open energy datasets.

A resource still needs users, evaluation criteria, and a boundary.

Six Seeds in Twelve Minutes

Generate before evaluating. Produce:

- two transfer ideas;
- two comparison or hidden-assumption ideas;
- one enabling-resource idea; and
- one wildcard idea driven only by your strongest curiosity.

Write each seed in two lines:

I noticed or learned _____.
I wonder whether/how _____.

Do not solve while generating

A seed may be rough. Capture distinct questions first; test feasibility afterward.

Label What You Know, Assume, and Need to Ask

For every promising seed, mark statements as:

Known
checked source or observation

Assumption
plausible but unverified

Question
requires evidence or advice

Useful transformation

"This method will help low-income households" becomes: "The paper reports [technical result]. I infer a possible household benefit, but must ask which outcome and whose experience would demonstrate it."

Turn a Seed into a Testable Research Question

Complete seven fields:

1. **Context:** where, when, and for whom?
2. **Target or outcome:** what will be measured?
3. **Method or factor:** what will be tested or compared?
4. **Evidence:** what data, observations, or sources are accessible?
5. **Baseline:** what simple alternative makes the result meaningful?
6. **Success rule:** what pattern or metric would answer the question?
7. **Boundary:** what will the project not claim?

Question frame

In [context], does [method/factor] improve or explain [outcome] compared with [baseline], using [evidence] under [boundary]?

If the Idea Uses Time Series, Specify the Clock

A time-series research question additionally needs:

- sampling frequency and time range;
- forecast horizon or event-detection window;
- information available at prediction time;
- chronological training and evaluation periods;
- relevant seasonal-naive baseline;
- events or groups for separate error analysis.

Time-series methods are optional

Use them only when temporal order is essential to the research question. The Week 18 evidence may inspire a stronger non-time-series project.

Feasibility Tournament: Choose Three Finalists

Score each seed from 1 (weak) to 5 (strong). Add evidence beside every score.

Criterion	Score	Evidence behind the score
Genuine curiosity	1–5	Would I still care after the first failed attempt?
Importance to a stakeholder	1–5	Who might use or be affected by the result?
Data/resource access now	1–5	Have I opened the data or confirmed access?
Testability	1–5	Is there a measurable outcome and baseline?
Small enough to finish	1–5	Can a minimum result be shown within three weeks?
Responsible boundary	1–5	Can I state what the result will not prove?

The total is a conversation aid, not an automatic decision. Keep the top three for feedback.

Mentor Speed Conference

Pitch each finalist in 45 seconds:

1. what sparked the idea;
2. the measurable question;
3. the evidence and baseline;
4. the smallest complete output; and
5. the largest uncertainty.

The mentor responds with only three items:

- one strength to preserve;
- one assumption to test within 72 hours;
- one way to shrink the scope.

Student chooses a **primary idea** and a **backup idea**; both remain revisable.

The Next Step Is Evidence, Not More Decoration

Choose one 72-hour evidence check for the primary idea:

- open the proposed dataset and verify columns, dates, and missingness;
- reproduce one small figure or baseline result;
- read two methods sections that define the intended measurement;
- ask one stakeholder or expert a precise question;
- test whether the necessary software and computation are available.

Decision rule

Write in advance what result means “continue,” “revise,” or “switch to the backup.”

Wrap-Up

Week 19 Homework

1. complete the no-AI time-series coding core;
2. compare persistence, seasonal naive, and lag-feature regression;
3. interpret errors by time rather than reporting only one metric;
4. expand the six idea seeds and score them with evidence;
5. perform one small reality check for the top two ideas;
6. submit Proposal Seed v0.2 with a primary and backup idea; and
7. publish a Feynman note explaining honest time-series evaluation.

The research deliverable is a decision with evidence

You are not expected to have a perfect final topic. You are expected to show why the next idea is more testable and feasible than the previous one.

Exit Ticket

Complete these statements:

1. A time-series feature leaks future information when . . .
2. The baseline I would try first for hourly load is . . . because . . .
3. One residual pattern that deserves investigation is . . .
4. The strongest research move I borrowed from Week 18 is . . .
5. My primary idea is currently stronger than my backup because . . .
6. Within 72 hours, I will test the assumption that . . .

Key Takeaways

1. Time order contains information and constrains honest evaluation.
2. Plot and audit timestamps before building a model.
3. Lags and rolling features create memory but must use only the past.
4. A model must beat relevant simple baselines on future data.
5. Error patterns can reveal both technical weaknesses and research questions.
6. Strong inspiration transfers a research move, not merely a topic title.
7. Generate alternatives before choosing; then use evidence, scope, and stakeholders to decide.

Read the past honestly. Choose the next question deliberately.

Next checkpoint: Proposal Seed v0.2, primary/backup decision, and one verified evidence path.